

6 Mammals produce ATP by anaerobic and aerobic respiration.

(a) The effect of exercise intensity on the lactate concentration in human blood was investigated.

- An athlete and a non-athlete performed the same exercise at the same intensity for three minutes.
- Their blood lactate concentrations were measured as soon as they finished exercising.
- This was repeated for different intensities of exercise.

(i) The diagram shows what happens to the lactate following exercise.



Which row shows the molecules in this diagram?

(1)

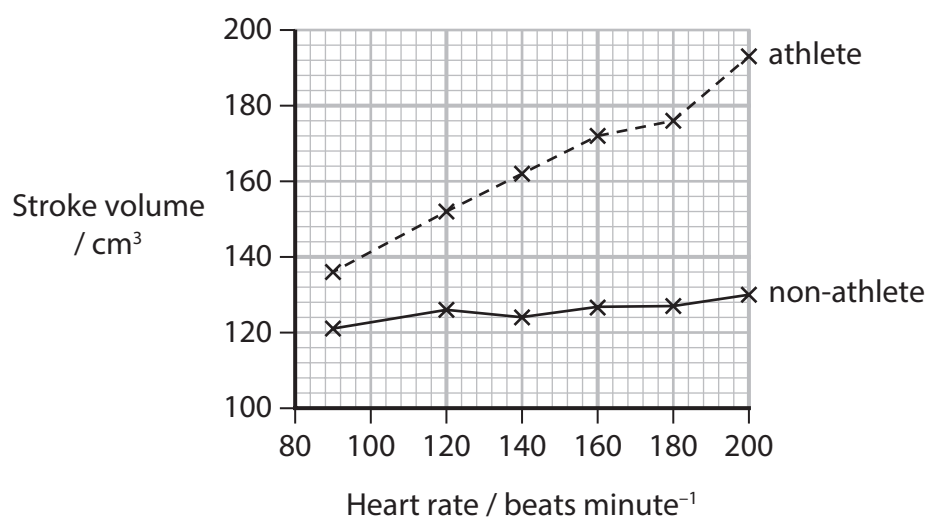
		Molecule W	Molecule Y	Molecule Z
☐	A	oxidised NAD	reduced NAD	pyruvate
☐	B	oxidised NAD	pyruvate	reduced NAD
☐	C	reduced NAD	oxidised NAD	pyruvate
☐	D	reduced NAD	pyruvate	oxidised NAD

*(ii) The table and graph show some of the results of this investigation.

Intensity of exercise / a.u.	Lactate concentration in blood / mmol dm^{-3}	
	Athlete	Non-athlete
0	1.0	1.5
60	1.1	2.2
120	1.4	3.7
180	3.2	6.2
240	6.4	10.0

Stroke volume is the volume of blood that leaves each ventricle during ventricular systole.

The graph shows the stroke volumes of the athlete and the non-athlete at different heart rates.



(6)

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(b) The photograph shows a coyote.



(Source: © Shaun Cunningham/Alamy Stock Photo)

Some coyotes have been poisoned with cyanide to prevent them from killing the livestock of farmers.

Cyanide inhibits cytochrome oxidase, the last carrier in the electron transport chain (ETC).

Explain how the action of cyanide would kill a coyote.

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(Total for Question 6 = 11 marks)